

AI Note: MCPSQLiteServer

Project Overview:

MCPSQLiteServer is a simple .NET project. It uses AI, SQLite, and MCP tools.

The project can:

- Send a question to Claude AI.
- Get an answer from Claude AI.
- Save messages in a SQLite database.
- Let MCP clients read table names, table schema, and SELECT query results.

Primary AI Features:

- The app connects to Claude AI.
- The user gives a prompt.
- Claude AI gives a text response.
- The app saves the user message and AI response in SQLite.
- The MCP tools help read the database safely.

The three MCP tools are:

- ``list_tables`` - shows all table names in the database.
- ``get_schema`` - shows the columns and data types of a table.
- ``run_select`` - runs only SELECT queries and shows the result.

AI Models:

This project uses Claude AI from Anthropic.

The model is selected using environment settings:

- ``ANTHROPIC_API_KEY`` is used to access Claude AI.
- ``ANTHROPIC_MODEL`` is used to choose the Claude model.

If no model is set, the project uses the default model from the app settings.

This project only uses text AI. It does not use image AI, voice AI, or local AI models.

Benefits:

- Easy to use and understand.
- Uses SQLite, so no separate database server is needed.
- Stores chat messages in a local database.
- MCP tools make it easy to read database data.
- ``run_select`` is safer because it only allows SELECT queries.
- API keys are stored in environment variables, not directly in code.

- The project can be improved later with more AI features.

Limitations:

- Claude AI needs an API key.
- Internet is needed to get AI responses.
- The current AI flow is simple.
- The app saves chat history, but it does not yet use old messages as memory.
- SQLite is good for small local apps, but not for large systems with many users.
- MCP tools only read data. They do not insert, update, or delete data.
- There is no login system.
- There is no frontend UI in this project.

Future AI Enhancements:

In the future, this project can add:

- Chat memory using old messages.
- Summary of long conversations.
- Search using AI embeddings.
- Ask questions in normal English and convert them to safe SQL.
- Better AI reports from database data.
- More model options.
- Error logs for AI requests.
- Data safety checks before sending text to AI.
- Local AI model support.
- Auto-generated database documentation.

Summary:

This project is a simple AI and SQLite project. It can talk to Claude AI, save messages, and let MCP clients read database data safely.

It is a good starting point for building AI tools with a local database.